Quantum Group Decision Making Method With Linguistic Z-Numbers Based on Trust Network

Shuli Yan, Yizhao Xu, Zaiwu Gong, and Francisco Javier Cabrerizo

***Abstract*—To address the fuzziness of information and the dynamic interference among trust paths in the uncertain social network group decision-making (SN-GDM) environment, this article evaluates information based on the linguistic Z-numbers (LZNs) representation and proposes a fuzzy group decision-making model considering quantum interference effects. First, the model updates trust relationships through a dynamic trust transfer process and incorporates quantum interference among trust paths to compute the comprehensive trust degree of each decision-maker (DM), which is then transformed into the corresponding DMs' weights. The interference angle is measured by the quantum gate transform and quantum Fourier transform (QFT). Second, an optimization model is constructed by integrating the distance relationship between DMs' psychological pReferences and evaluation information to determine the optimal attributes' weights. On this basis, the scores of the alternatives are calculated and ranked. Finally, a case study on emergency response is conducted to verify the feasibility and effectiveness of the proposed model, followed by a sensitivity analysis of the rotation angle parameter and the information transmission discount factor, as well as a comparative analysis with existing decision-making models. The results demonstrate that the proposed model not only possesses strong applicability and computational efficiency but also accurately characterizes the progressive transmission process of trust relationships and enables efficient measurement of quantum interference effects.**

***Index Terms*—Linguistic Z-numbers (LZNs), quantum gate transformation, quantum interference effect, social network group decision-making (SN-GDM), trust relationship.**

NOMENCLATURE

D	Set of DM $D = \{d_1, d_2, \ldots, d_s\}$.
A	Set of alternatives $A = \{a_1, a_2, \ldots, a_m\}$.
t_{kh}	Trust value from d_k to d_h, $t_{kh} \in \{0, 1\}$.
T	Trust matrix, $T = (t_{kh})_{s\times s}$.
COD_h	Comprehensive relative out-degree centrality of d_h.
δ_{kh}	Interference angle between d_k and d_h.
$\langle \upsilon_k'' \vert \vartheta_h'' \rangle$	Inner product between υ_k'' and ϑ_h''.
H	Hadamard gate matrix.
$p'(d_k)$	Comprehensive trust level of d_k.
Q_i	Score value of a_i.
ρ_5	Discount factor, $\rho_5 \in [0, 1]$.
$\varepsilon_i (i = 1, 2)$	Regularized parameter.
e_k, e_h	Trust edge between DM d_k and d_h.
$\tilde{Z}^k = (\tilde{z}_{ij}^k)_{m\times n}$	Linguistic scale matrix of d_k.
$p(d_k)$	Comprehensive relative out-degree centrality of d_k.
TID_h	Trust input degree of d_h.
z_0^{k+}	Best evaluation value of ideal solution for d_k.
ϑ_h	Preference vector of d_h.
$R_y(\theta)$	Rotation operation matrix.
$p(d_k \vert d_h)$	Conditional probability in QLBN.
ρ_3	Resolution coefficient, $\rho_3 \in [0, 1]$.
ρ_6	Adjustment parameter, $\rho_6 \in [0, 1]$.
C	Set of attributes $C = \{c_1, c_2, \ldots, c_n\}$.
$Z^k = (z_{ij}^k)_{m\times n}$	Initial evaluation matrix of d_k.
$\hat{Z}^k = (\hat{z}_{ij}^k)_{m\times n}$	Decision matrix of d_k.
ω_j^k	Weight of attribute c_j for d_k.
TOD_h	Trust output degree of d_h.
υ_k	Preference vector of d_k.
$\Vert \cdot \Vert$	Normalization of preference vector.
QFT	Quantum Fourier transform.
λ_k	Weight of d_k.
ρ_4	Punishment coefficient, $\rho_4 \geq 0$.
$F_{1,2}^*(\alpha)$	Linguistic scale function.

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I. INTRODUCTION

IN A complex decision-making environment with ambiguous information, dynamic changes, and numerous stakeholders, the decision-making process is confronted with numerous uncertain factors [1]. Especially in the process of multiattribute group decision-making (MAGDM), the knowledge limitations and subjective biases of decision-makers (DMs), coupled with the difficulty in obtaining accurate quantitative data, often affect the quality of decision-making [2], [3]. In this context, linguistic information has gradually attracted attention as an important tool for dealing with uncertainty [4]. Since the linguistic Z-numbers (LZNs) [5] not only describe fuzzy information in linguistic form but also integrate reliability components, it provides a fuzzy quantization range and credibility expression [7]. Therefore, in this article, LZNs are adopted as the information representation to describe the characteristics of information [8], [9] in complex and uncertain decision-making problems more comprehensively and accurately.

In the context of MAGDM, social networks (SNs) not only provide convenient channels for information dissemination, but also significantly influence the decision-making process through their complex structures and interaction patterns [10], [11]. Trust paths may influence each other, enhancing or weakening each other's effects. This nonlinear interference has a significant impact on decision-making outcomes. However, traditional social network group decision-making (SN-GDM) models usually assume that trust paths are independent of each other [12], [13]. This assumption fails to reflect the mutual interference effects between paths and thus appears no longer applicable. Furthermore, the trust relationship itself is dynamically evolving [14], [15]. Over time, the structure of SN and the interaction patterns among DMs may change, and these changes will directly affect the renewal of trust and its effect on decision outcomes [16], [17]. For instance, the trust relationship between nodes may constantly adjust as interactions and information flows change, and alterations in the network structure may also lead to a redistribution of influence among DMs [18]. Therefore, in SN-GDM, the mutual interference among trust paths and the dynamic evolution of trust relationships must be taken into account to more accurately reflect the complex interactions and relationships among DMs.

In MAGDM problems, reasonably determining the attributes' weights of alternatives is of great significance for ensuring the accuracy of decisions. In the existing literature, a variety of methods for calculating attribute weights have been proposed, including the entropy weight method, analytic hierarchy process (AHP), and fuzzy set-based weight determination method [19], [20], [21]. However, these methods still have limitations in the dynamic adjustment of attribute weights in complex decision-making environments. To balance the objectivity of weight distribution [22] and the psychological perception characteristics of DMs [23], the objective evaluation mechanism of TOPSIS can be combined with the behavioral perception modeling of TODIM, thereby simultaneously enhancing the rationality, adaptability, and interpretability of weight determination in the dynamic decision-making process.

In the traditional group decision models, it is assumed that each DM evaluates decision information, and there is no interference between DMs. In fact, the opinions and choices of DMs are often significantly influenced by the opinions and emotions of others, and the opinion deviation caused by the interaction and influence of DMs produces the effect of group interference [24], [25]. Quantum probability theory can simulate such interactions in a quantum-like Bayesian network (QLBN) and more realistically describe and predict human group decision-making behavior [26]. When describing the dynamics of group decision making, especially when considering the complex psychological and social interactions among DMs, quantum models can reveal the paradoxes, suboptimal choices, and irrational behaviors in human decision making [27], [28], [29]. The quantum interference effect has a significant effect on the calculation of the final probability, thus affecting the decision result [30], [31]. In quantum decision theory, interference values are usually represented by phase angles, but these angles often have uncertainty [32], [33]. For this reason, existing studies have proposed an interference angle measurement method based on confidence entropy and vector angle [34], [35].

Although existing research has made many advancements in the field of MAGDM, there are still several key issues in dealing with complex and uncertain SN-GDM environments. The research motivation of this article mainly focuses on the following points:

1) Most of the existing SN-GDM models assume that trust paths are independent of each other, which makes it impossible for them to capture the nonlinear interference effects between trust relationships.

2) The existing interference quantization methods either fail to accurately represent the direction and intensity of interference between paths or overly depend on global information. Meanwhile, these methods usually lack physical interpretability, thus making it difficult to conduct in-depth analysis of the complex social interactions and information transmission processes among DMs.

3) The existing methods either rely on objective data while ignoring the cognitive preferences of DMs, or use subjective methods that are easily influenced by personal biases, resulting in unreasonable weight distribution. However, in SNs, the psychological perception differences of DMs and their varying degrees of preference for information can significantly affect the final decision-making outcome.

On this basis, this article proposes a social trust network-based multiattribute quantum group decision-making model (STN-MAQGDM), which incorporates LZNs for evaluation information representation and aims to more faithfully characterize the interactions among DMs and their impacts on decision-making outcomes through the dynamic adjustment of trust relationships and quantum interference effects. The main contributions of this article are as follows.

1) A multiattribute quantum group decision-making model based on a trust network is proposed. This model innovatively considers the dynamic adjustment mechanism of two-stage trust relationships and introduces quantum interference effects to capture the interaction between trust paths, thereby more accurately describing the transfer process of trust relationships.

2) An interference angle measurement method based on quantum gate transform and quantum Fourier transform (QFT) is proposed, which can effectively capture the nonlinear superposition between paths and improve the accuracy of interference effect measurement.

3) An attributes' weights optimization model based on the TOPSIS-TODIM idea is constructed, which can simultaneously integrate objective evaluation information and the

psychological perception differences of DMs in the dynamic SN decision-making environment.

The article is organized as follows. Section II reviews LZNs, trust networks, and the quantum probability theory framework. In Section III, we introduce the LZNs multiattribute quantum group consensus decision model based on trust networks, which combines trust networks, quantum probability, and Bayesian networks, detailing the decision steps. Section IV verifies the STN-MAQGDM through a case study of meteorological disaster emergency rescue and validates the model's stability and reliability through simulation, sensitivity, and comparative analyses. Finally, Section V summarizes the main findings and contributions of this article and suggests directions for future research.

II. Preliminaries

A. Linguistic Z-Numbers

Zadeh [6] proposed the Z-numbers in 2011, which are used to describe the uncertain information composed of restrictive information and its confidence level. With the deepening of research, the Z-numbers have been further expanded into the LZNs [8], [9], [38], that is, both of its components are represented in the form of linguistic terms to deal with ambiguous, subjective, and difficult-to-quantify decision-making information more effectively.

Definition 1: [6]: Let X be the domain of discourse, $L_1 = \{l_0, l_1, \ldots, l_{2g}\}$ and $L_2 = \{l'_0, l'_1, \ldots, l'_{2g'}\}$ be two finite and totally ordered sets of linguistic terms, where g and g' are nonnegative integers. In addition, let $A_{\varphi(x)} \in L_1$ and $B_{\phi(x')} \in L_2$. The set of LZNs Z in X consists of objects of the following form:

$$Z = \left\{ \left(x, A_{\varphi(x)}, B_{\varphi(x')}\right) | x \in X \right\}. \quad (1)$$

The set Z is defined by two linguistic terms, $A_{\varphi(x)}$ and $B_{\varphi(x')}$. Here, $A_{\varphi(x)}$ denotes a fuzzy limit on the allowable range of values for uncertain variables, while $B_{\varphi(x')}$ measures the reliability of these uncertain variables. In addition, the linguistic term sets L_1 and L_2 differ from each other, representing distinct preference information, respectively.

When X has only one element, the set of LZNs simplifies to $(A_{\phi(x)}, B_{\varphi(x')})$. For convenience, this LZN is expressed as $z = (A_{\varphi(x)}, B_{\varphi(x')})$, where $A_{\varphi(x)} \in L_1$ and $B_{\phi(x')} \in L_2$ are two linguistic terms.

B. Linguistic Scale Function

When making decisions with linguistic terms, it is often impossible to directly compare and calculate. Therefore, in order to make the information of decision-making more scientific and effective and the decision-making process more convenient, a quantitative method is needed to convert linguistic decisions into specific values.

Definition 2: [25]: Let the linguistic term set be denoted as $S = \{s_\alpha \mid \alpha = 0, 1, 2, \ldots, 2m\}$, where s_α represents a finite set of linguistic terms defined over an ordinal scale and the terms are arranged in increasing order of semantic preference. Suppose there is a set of values $\theta_\alpha \in [0, 1]$, and $\theta_\alpha (\alpha = 0, 1, 2, \ldots, 2m)$, the mapping $F^*(\theta) \in [0, 1]$ from s_α to $\theta_\alpha (\alpha = 0, 1, 2, \ldots, 2m)$ is defined as follows:

$$F^* : s_\alpha \to \theta_\alpha \, (\alpha = 0, 1, 2, \ldots, 2m). \quad (2)$$

θ_α is strictly monotonically increasing, meaning that for any $\alpha = 0, 1, 2, \ldots, 2m$, $0 \le \theta_0 \le \theta_1 \le \cdots \le \theta_{2m} \le 1$. θ_α can be regarded as a preference in the DM's linguistic: the larger the value of θ_α, the better the corresponding s_α. In addition, the linguistic scale function (LSF) is strictly monotonically increasing with the increase of s_α.

The traditional LSF [38] often shows insufficient gradient in the middle or compression in the low-value area in numerical distribution. It is difficult to effectively distinguish the subtle differences between neutral terms and low-intensity terms, which limits the fine expression ability of fuzzy linguistic information. On this basis, this article proposes two types of improved LSF1 and LSF2 to enhance the numerical resolution of linguistic terms in different semantic intervals.

Definition 3: LSF1 is defined as

$$F_1^*(\alpha) = \begin{cases} \dfrac{\ln\left(1 + \left(\frac{\alpha}{m}\right)^{\beta_1} + \left(\frac{\alpha}{m}\right)^{\beta_2}\right)}{2\ln 3}, \alpha = 0, 1, \ldots, m \\ 1 - \dfrac{\ln\left(1 + \left(\frac{2m-\alpha}{m}\right)^{\beta_1} + \left(\frac{2m-\alpha}{m}\right)^{\beta_2}\right)}{2\ln 3} \\ \alpha = m+1, m+2, \ldots, 2m \end{cases} \quad (3)$$

where $\beta_1, \beta_2 \in (0, 2)$.

LSF1 is designed to address the rigidity of traditional linear or symmetric scale functions by introducing a logarithmic form with two tunable exponents, β_1 and $\beta_2 (\in (0, 2))$. This dual-parameter structure enables flexible adjustment of the growth rate on both sides of the scale, allowing the function to better simulate asymmetric or nonlinear shifts in linguistic preferences. Compared with convex or concave power functions, LSF1 provides greater modeling adaptability and finer control over local sensitivity, making it more suitable for capturing gradual and imbalanced preference transitions in uncertain environments.

Definition 4: LSF2 is defined as

$$F_2^*(\alpha) = \frac{1}{2}\left(1 - \cos\left(\frac{\alpha\pi}{2m}\right)\right), \quad \alpha = 0, 1, 2, \ldots, 2m. \quad (4)$$

LSF2 adopts a cosine-based trigonometric form, which yields a characteristic "steep-middle and smooth-ends" distribution. This structure is particularly effective for enhancing the resolution of intermediate linguistic terms, where traditional functions often suffer from flat gradients or value crowding. By amplifying the distinctions between neutral and adjacent terms, LSF2 improves the semantic granularity of fuzzy evaluations and strengthens the interpretability of mid-scale linguistic mappings.

Theorem 1: The LSF $F_{1,2}^*(\alpha)$ satisfies the following three properties.

1) $F_{1,2}^*(0) = 0$, $F_{1,2}^*(m) = 1/2$, $F_{1,2}^*(2m) = 1$.
2) $F_{1,2}^*(\alpha)$ in $\alpha = 0, 1, 2, \ldots, 2m$ is monotonically increasing.
3) For any $F_{1,2}^*(i)$ and $F_{1,2}^*(j)$, when $i + j = 2m$, $F_{1,2}^*(i) + F_{1,2}^*(j) = 1$.

The proof of Theorem 1 is provided in Appendix A.

In summary, the two LSFs proposed in this article satisfy the three basic properties of theorem 1. To verify the superiority of the LSF proposed in this article, assuming $m = 5$, this article makes a comparison with the traditional LSF $H_o(\alpha)(o = 1, 2, 3, 4)$ [38], as shown in Fig. 1.

According to Fig. 1, traditional LSF like H_2 yield values below 0.05 for terms s_0 to s_4, indicating severe compression in the low-end region. Similarly, H_3 grows too rapidly at the beginning (e.g., $H_3(1) \approx 0.63$), which leads to value crowding in the high-end. In contrast, LSF1 provides steeper value changes at both ends (e.g., $F_1^*(1) \approx 0.19, F_1^*(9) \approx 0.95$), while LSF2 exhibits a prominent increase in the middle region ($F_2^*(4) = 0.35, F_2^*(6) = 0.65$). Therefore, the LSF proposed in this article has a stronger semantic distinction between neutral and extreme terms and is more effective in the quantification of fuzzy linguistic terms in the collection of uncertain decision-making processes.

Definition 5: [38]: The weighted Euclidean distance between $z_x = (A_{\varphi(x)}, B_{\varphi(x)})$ and $z_{x'} = (A_{\varphi(x')}, B_{\varphi(x')})$ is defined as follows:

$$d\left(z_x, z_{x'}\right) = \sqrt{\begin{array}{l}\rho_1\left(F_1^*\left(A_{\varphi(x)}\right) - F_1^*\left(A_{\varphi(x')}\right)\right)^2 \\ +\rho_2\left(F_2^*\left(B_{\phi(x)}\right) - F_2^*\left(B_{\phi(x')}\right)\right)^2\end{array}} \quad (5)$$

where $\rho_1 \in [0, 1]$ and $\rho_2 \in [0, 1]$ are parameters that reflect the importance of A and B in the LZN, and $\rho_1 + \rho_2 = 1$.

C. Trust Network

A trust network [29] is a directed graph $G(D, E)$, where $D = \{d_1, d_2, \ldots, d_s\}$ represents the set of DMs, and $s \geq 3$ is the number of DMs. The set of directed edges $E = \{e_1, e_2, \ldots, e_q\}$ describes the direct influence relationships between the DMs. Specifically, an edge e_k indicates that d_k directly influences d_h.

To quantify the trust relationships between DMs, a sociometric matrix $T = (t_{kh})_{s\times s}$ is introduced to describe the relationships between the nodes in the trust network. Each element t_{kh} in the matrix represents whether d_k directly influences d_h, and is defined as follows:

$$t_{ij} = \begin{cases}1, & (d_k, d_h) \in E \\ 0, & \text{other}\end{cases}$$
$$i = 1, 2, \ldots, s, \quad j = 1, 2, \ldots, s, i \neq j. \quad (6)$$

D. Quantum Probability Theory

In quantum probability theory, the occurrence probability of an event is no longer determined by the classical additive principle but by the inner product of probability amplitudes between quantum states [24]. Suppose the state transition $P \to A$ can occur through intermediate states G and B; the total probability amplitude is given by

$$\langle A|P\rangle = \langle G|P\rangle \langle A|G\rangle + \langle B|P\rangle \langle A|B\rangle . \quad (7)$$

The corresponding event probability is expressed as

$$\begin{aligned}p\left(A\right) &= |\langle A|P\rangle|^2 \\ &= p\left(G\right)p\left(A|G\right) + p\left(B\right)p\left(A|B\right) \\ &\quad + 2\sqrt{p\left(G\right)p\left(A|G\right)p\left(B\right)p\left(A|B\right)}\cos\left(\alpha-\beta\right) \quad (8)\end{aligned}$$

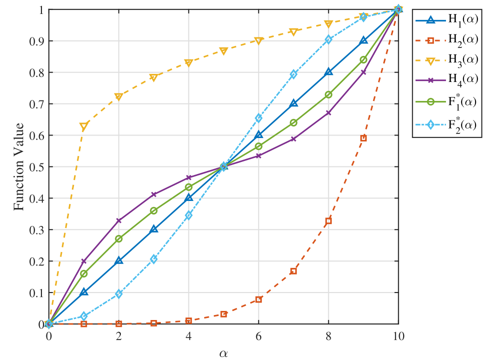

Fig. 1. Image shape of each LSF when $m = 5$.

where $\cos(\alpha - \beta)$ represents the interference term, which captures the superposition effect between different cognitive paths. When the interference term equals zero, the quantum model degenerates into the classical probability model.

III. SN Trust Model Considering Quantum Interference Perspective

To facilitate understanding of the meanings of the variables in this article, the Nomenclature provides a unified explanation of the main variable symbols.

A. Collect Decision Information

1) Evaluation information collection. Each DM $d_k(k = 1, 2, \ldots, s)$ uses LZN $z_{ij}^k = (A_{ij}^k, B_{ij}^k)(i = 1, 2, \ldots, m, j = 1, 2, \ldots, n, k = 1, 2, \ldots, s)$ to represent the evaluation information of alternative $a_i(i = 1, 2, \ldots, m)$ about attribute $c_j(j = 1, 2, \ldots, n)$, and the evaluation information matrix is expressed as $Z^k = (z_{ij}^k)_{m\times n}(k = 1, 2, \ldots, s)$, that is,

$$Z^k = \begin{pmatrix} z_{11}^k & z_{12}^k & \cdots & z_{1n}^k \\ z_{21}^k & z_{22}^k & \cdots & z_{2n}^k \\ \vdots & \vdots & \ddots & \vdots \\ z_{m1}^k & z_{m2}^k & \cdots & z_{mn}^k \end{pmatrix}, \quad k = 1, 2, \ldots, s. \quad (9)$$

On this basis, the evaluation information matrix $Z^k = (z_{ij}^k)_{m\times n}(k = 1, 2, \ldots, s)$ can be transformed into the LSF matrix $\tilde{Z}^k = (\tilde{z}_{ij}^k)_{m\times n} = \left[F_1^*(A_{ij}^k), F_2^*(B_{ij}^k)\right]_{m\times n}(k = 1, 2, \ldots, s)$ by (3) and (4) through the LSF.

2) In SN-GDM analysis, dealing with the uncertainty and ambiguity of linguistic terms is a key problem. Traditional numerical methods often struggle to effectively capture the fuzziness of linguistic terms. However, gray system theory offers an effective method to address uncertainty and incomplete information [39], [40]. This article uses the gray relational degree to represent LZNs, converting them into crisp numbers. To prevent information loss, the relative distance between decision-making units and ideal solutions is described by the gray relational degree. By reflecting the proximity of $\tilde{z}_{ij}^k(i = 1, 2, \ldots, m, j = 1, 2, \ldots, n, k = 1, 2, \ldots, s)$ to ideal

and nonideal solutions, the LZNs are transformed into a crisp number, as follows:

$$\begin{aligned}\hat{z}_{ij}^{k} &= \xi\left(\tilde{z}_{j}^{k+}, \tilde{z}_{ij}^{k}\right)\\ &= \frac{\min_k \min_i d\left(\tilde{z}_{ij}^{k}, \tilde{z}_{0}^{k+}\right) + \rho_3 \max_k \max_i d\left(\tilde{z}_{ij}^{k}, \tilde{z}_{0}^{k+}\right)}{d\left(\tilde{z}_{ij}^{k}, \tilde{z}_{0}^{k+}\right) + \rho_3 \max_k \max_i d\left(\tilde{z}_{ij}^{k}, \tilde{z}_{0}^{k+}\right)}\\ & i = 1,2,\ldots,m,\ \ j = 1,2,\ldots,n,\ \ k = 1,2,\ldots,s \end{aligned} \tag{10}$$

where $\rho_3 \in (0,1)$ is the resolution coefficient, $\tilde{Z}_0^{k+} = (\tilde{z}_1^{k+}, \tilde{z}_2^{k+}, \ldots, \tilde{z}_n^{k+})$ is the optimal value of the evaluation matrix given by the kth DM, and $\tilde{z}_j^{k+} = \max_i \tilde{z}_{ij}^{k} (i = 1,2,\ldots,m, j = 1,2,\ldots,n)$.

Theorem 2: Gray relational degree $\hat{z}_{ij}^{k} = \xi(\tilde{z}_j^{k+}, \tilde{z}_{ij}^{k})$ has the following properties.

1) The value range of $\hat{z}_{ij}^{k}$ is between $(0,1]$. For any of the i, j, k there are: $0 < \hat{z}_{ij}^{k} \le 1$.

2) The smaller the distance between the LZN and the optimal value, the greater the gray correlation degree. That is, when $d(\tilde{z}_{ij}^{k}, \tilde{z}_0^{k+}) < d(\tilde{z}_{mn}^{k}, \tilde{z}_0^{k+})$, then $\hat{z}_{ij}^{k} > \hat{z}_{mn}^{k}$.

3) The correlation degree $\hat{z}_{ij}^{k}$ increases monotonically with the resolution coefficient $\rho_3 \in (0,1)$.

The proof of Theorem 2 is provided in Appendix B.

To verify the rationality and effectiveness of the LZNs clarification method proposed in this article, a comparative analysis is conducted against three representative methods from a methodological perspective. 1) LZN Method 1 [41] is based on a traditional LSF, which constructs crisp values by linearly multiplying the linguistic term score with the confidence weight. Although simple to compute, it ignores the nonlinear structure and semantic asymmetry between the two components, which often leads to information compression. 2) LZN Method 2 [38] introduces a distance measure and constructs a relative closeness degree to enhance discriminative ability, but it still relies on a classical scale function and fails to adequately capture the interaction between linguistic fuzziness and confidence. 3) Although LZN Method 3 [39] also adopts the idea of gray relational degree, it requires the calculation of both positive and negative ideal solutions, resulting in a more complex structure and higher computational cost; moreover, introducing a negative ideal solution may lead to information perturbation.

In contrast, the method proposed in this article introduces two improved LSFs and completes the clarification process by combining the gray relational degree with respect to the positive ideal solution. This approach maintains structural rationality and computational efficiency while providing stronger numerical discrimination.

B. DM's Weight Determination Based on Trust Network

In trust networks, since traditional weighting methods are mostly based on static trust values or the assumption of path independence, it is difficult to reflect the dynamic evolution and interaction effects among multiple paths in the trust network [37]. To evaluate the trust level of DMs more accurately, this section first constructs a trust network optimization model and obtains the modified trust relationship through a dynamic adjustment mechanism; On this basis, the quantum interference between trust paths is considered, and an interference angle measurement method based on quantum gates and Fourier transform is proposed. Phase information is introduced to improve the quantization accuracy of the interference effect. Finally, through the aggregation of trust paths, the DMs' weights are scientifically determined. Therefore, this method is more in line with the true characteristics of the trust network structure in an uncertain environment.

1) Comprehensive Trust Model Considering Trust Path and Centrality: In SN-GDM, trust relationships usually stem from the interactions and historical behaviors among members and can be quantified through ratings or cooperation records. To express different degrees of trust in more detail, the trust value is expanded to any real number within the interval $[0,1]$ [42]. The trust matrix $T = (t_{kh})_{s\times s}(t_{kh} \in [0,1])$ is used to represent the degree of trust between DMs. To effectively calculate and optimize these trust relationships, this article constructs a model based on the maximum trust path and centrality, enabling the simulation and determination of the group's optimal trust relationship.

First, the degree centrality of each DM is calculated according to the initial trust matrix $T = (t_{kh})_{s\times s}(t_{kh} \in [0,1])$, and the comprehensive relative out-degree centrality COD_h of DM d_h is defined as [11]

$$\mathrm{COD}_h = \frac{\mathrm{TOD}_h \times \mathrm{TID}_h}{\max\left(\mathrm{TOD}_h \times \mathrm{TID}_h\right)}, h = 1,2,\ldots,s. \tag{11}$$

The larger the comprehensive relative out-degree centrality COD_h is, the more likely d_h is to trust other DMs. Trust input degree $\mathrm{TID}_h (h = 1,2,\ldots,s)$ represents the sum of the trust degree of other nodes to the trusted path of this node, and trust output degree $\mathrm{TOD}_h (h = 1,2,\ldots,s)$ represents the sum of the trust degree of this node to the trusted path of other nodes

$$\mathrm{TID}_h = \sum_{k=1,k\neq h}^{s} t_{kh}, \quad h = 1,2,\ldots,s \tag{12}$$

$$\mathrm{TOD}_h = \sum_{h=1,k\neq h}^{s} t_{kh}, \quad h = 1,2,\ldots,s. \tag{13}$$

Second, consider the two-stage trust path. On the basis of obtaining primary and secondary trust relationships, the top DM organizes DMs to communicate with each other and deepen mutual understanding and knowledge. A trust network optimization model [see (14)] is then built to optimize trust values by dynamically adjusting trust relationships between nodes, thereby improving the reliability and effectiveness of the entire trust network. The model dynamically adjusts the trust values to reflect changes in trust during actual communication and to avoid decision errors caused by excessively high trust values. In addition, the model introduces transitive constraints to reasonably simulate the influence of indirect trust transfer and ensures the robustness of the trust relationship during the optimization process by setting an upper limit on trust values and incorporating a penalty mechanism. The optimization model is as follows:

$$M_1 : \max \left\{ \begin{array}{l} \sum_{\mathrm{h}=1}^{s} \left(\frac{\mathrm{TOD}_h + \mathrm{TID}_h}{2}\right) - \\ \rho_4 \sum_{h=1}^{s}\sum_{k=1}^{s} \max\left(0, {t'}_{kh} - (t_{kh}+\varepsilon_1)\right)^2 \end{array} \right\}$$

$$\text{s.t.} \begin{cases} t_{kh} \le {t'}_{kh} \le t_{kh} + \varepsilon_2 & (14-1) \\ {t'}_{kh} \ge \rho_5 \cdot \left(t_{kh} + \rho_6 \sum_{q=1, q\ne h}^{s} t_{hq} t_{\mathrm{qk}} \right) & (14-2) \\ \mathrm{TID}_h = \sum_{k=1,k\ne h}^{s} {t'}_{kh} & (14-3) \\ \mathrm{TOD}_h = \sum_{h=1,k\ne h}^{s} {t'}_{kh} & (14-4) \\ 0 \le {t'}_{kh} \le 1 & (14-5) \\ \rho_4 \ge 0, 0 \le \rho_i \le 1,\ \ i = 5,6 & (14-6) \\ \varepsilon_i \ge 0,\ \ i = 1,2 & (14-7) \\ k = 1,2,\ldots,s,\ \ h = 1,2,\ldots,s & \\ q = 1,2,\ldots,s,\ \ q \ne h \ne k & (14-8) \end{cases} \tag{14}$$

where $\rho_4(\rho_4 \ge 0)$ represents the punishment coefficient, which is used to control the intensity of punishment. $\rho_5 \in [0, 1]$ is an information transmission discount factor that reflects the attenuation effect during trust transfer, where a smaller value of ρ_5 indicates a faster decay of trust in the transmission process. $\rho_6 \in [0, 1]$ is an adjustment parameter used to control the degree of influence of indirect trust on the final trust value. A larger $\rho_6 \in [0, 1]$ means more indirect trust, a smaller $\rho_6 \in [0, 1]$ means less indirect trust, and $\varepsilon_i (i = 1, 2)$ is a regularized parameter.

Theorem 3: Model M_1 has an optimal solution.

The proof of Theorem 3 is provided in Appendix C.

The objective function of the model is to maximize the comprehensive trust of each node, while introducing a penalty term to maintain the model's realistic rationality. Constraint (14-1) ensures that the optimization process does not reduce the initial trust between nodes. Constraint (14-2) considers indirect trust, with trust gradually diminishing during the transfer process. Constraints (14-3) and (14-4) pertain to trust calculations, while the remaining constraints define the range of each parameter interval.

On this basis, the second stage trust matrix $T' = ({t'}_{kh})_{s\times s}, {t'}_{kh} \in [0, 1]$ based on trust transfer is obtained by solving model M_1.

2) Comprehensive Trust Level Calculation Considering Quantum Interference: In the decision-making process, the opinions of DMs are often influenced by the relationship of trust. Within trust networks, it is considered that there is an interference effect between each path, which impacts the degree of trust between DMs. The quantum interference effect between each trust relationship path refers to the interaction of trust information transmitted through different paths, altering the degree of trust that DMs place on other nodes. Therefore, quantum probability calculations can be performed based on QLBN.

First, the comprehensive relative out-degree centrality of each DM constitutes the first layer probability of QLBN

$$p(d_h) = \frac{\mathrm{COD}_h}{\sum_{h=1}^{s} \mathrm{COD}_h}, \quad h = 1, 2, \ldots, s. \tag{15}$$

Second, the trust path weights between DMs are used as conditional probabilities

$$p(d_k|d_h) = \frac{{t'}_{hk}}{\sum_{k=1,k\ne h}^{s} {t'}_{hk}}$$
$$i = 1, 2, \ldots, m,\ \ k = 1, 2, \ldots, s. \tag{16}$$

Finally, to calculate the comprehensive trust level of each DM

$$p'(d_k) = \sum_{h=1,h\ne k}^{s} p(d_h)\, p(d_k|d_h) + R_{d_k}, \quad k = 1, 2, \ldots, s \tag{17}$$

where the interference term R_{d_k} is defined as

$$R_{d_k} = 2 \sum_{h=1,h\ne k}^{s-1} \sum_{h'=h+1}^{s} \sqrt{p(d_h)\, p(d_k|d_h)} \times \sqrt{p(d_{h'})\, p(d_k|d_{h'})} \cos\delta_{hh'}, \quad k = 1, 2, \ldots, s. \tag{18}$$

Algorithm 1 Interference Angle Calculation Based on Quantum Gate Transformation

Input: Preference vectors υ_k and ϑ_h, dimension $s - 1$, rotation angle θ.
Output: Interference angle δ_{kh}.
Begin
  Step 1: Normalize preference vectors using Eq. (21).
  Step 2: Define Hadamard gate H and rotation operator $R_y(\theta)$.
  Step 3: Apply H and $R_y(\theta)$ to each element:
  for $i = 1$ **to** $s - 1$ **do**
    Apply Hadamard gate H;
    Apply rotation operator $R_y(\theta)$.
  end
  Step 4: Perform the quantum Fourier transform using Eq. (26).
  Step 5: Compute the inner product via Eq. (27) and obtain δ_{kh} using Eq. (29).
End

3) Interference Angle Measure Based on Quantum Gate Transformation: In a complex SN-GDM environment, trust relationships among DMs are usually transmitted through multiple channels, which may reinforce or weaken each other. This phenomenon can be, respectively, regarded as constructive interference and destructive interference. To describe this nonlinear relationship, this article adopts the idea of quantum probability theory and uses quantum gate operations to simulate the superposition and coherence of trust paths, thereby overcoming the limitations of classical similarity measures. The detailed procedure of the proposed method is presented in Algorithm 1.

Specifically, the preference vectors are processed and analyzed by using quantum gates [44] and the QFT [43]. By applying the Hadamard transform, the quantum state enters the superposition state, and more data features are captured by introducing phase information. Further adjust the phase of the quantum state through rotational transformation. Finally, the quantum state is transformed into the frequency domain through the QFT to capture the complex interrelationships and coherence, as shown in Fig. 2.

The upper and lower quantum lines in Fig. 2 represent the trust propagation vectors $|\upsilon_k\rangle$ and $|\vartheta_h\rangle$, respectively. The Hadamard gates H generate superposition states that represent multiple trust propagation paths. Rotation gates $R_y(+\theta)$ and $R_y(-\theta)$ introduce opposite phase shifts to simulate constructive and destructive interference. The QFT maps the quantum states into the frequency domain to capture nonlinear relationships. Finally, the measurement of the inner product $\langle v_k''|\vartheta_h''\rangle$ yields the interference angle δ_{kh}. The calculation process is as follows.

1) Construct preference vector $\upsilon_k(k = 1, 2, \ldots, s)$ and $\vartheta_h(h = 1, 2, \ldots, s)$

$$\begin{aligned}\upsilon_k &= \left[\, p(d_h)p(d_1|d_h), p(d_h)p(d_2|d_h), \ldots, p(d_h)p(d_k|d_h)\,\right]^T\\ \vartheta_h &= \left[\, p(d_k)p(d_1|d_k), p(d_k)p(d_2|d_k), \ldots, p(d_k)p(d_h|d_k)\,\right]^T\\ &k = 1, 2, \ldots, s,\ \ h = 1, 2, \ldots, s,\ \ h \neq k \qquad (19)\end{aligned}$$

where

$$\begin{aligned}\upsilon_{kh} &= p(d_k)p(d_h|d_k), \vartheta_{kh} = p(d_k)p(d_h|d_k)\\ &k = 1, 2, \ldots, s,\ \ h = 1, 2, \ldots, s,\ \ h \neq k. \qquad (20)\end{aligned}$$

2) *Vector Normalization:* Normalize two column vectors of the same dimension to quantum states

$$\begin{aligned}|\upsilon_k\rangle &= \frac{1}{\|\upsilon_k\|}(\upsilon_{1h}, \upsilon_{2h}, \ldots, \upsilon_{kh})^T\\ |\vartheta_h\rangle &= \frac{1}{\|\vartheta_h\|}(\vartheta_{k1}, \vartheta_{k2}, \ldots, \vartheta_{kh})^T\\ &h, k = 1, 2, \ldots, s,\ \ h \neq k. \qquad (21)\end{aligned}$$

3) Apply Hadamard gate H and rotation operators $R_y(\theta)$ for each qubit

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1\\ 1 & -1\end{pmatrix} \qquad (22)$$

$$R_y(\theta) = \begin{pmatrix}\cos(\theta/2) & -\sin(\theta/2)\\ \sin(\theta/2) & \cos(\theta/2)\end{pmatrix}. \qquad (23)$$

Considering the relative phase difference between two trust propagation paths, the rotation transformation is further extended to the $s-1$ dimensional case by applying opposite rotation directions, $+\theta$ and $-\theta$, to represent constructive and destructive interference, as follows:

$$\begin{aligned}\left|\upsilon_k''\right\rangle &= R_y(+\theta)^{\otimes(s-1)} H^{\otimes(s-1)} |\upsilon_k\rangle\\ \left|\vartheta_h''\right\rangle &= R_y(-\theta)^{\otimes(s-1)} H^{\otimes(s-1)} |\vartheta_h\rangle\\ &k = 1, 2, \ldots, s,\ \ h = 1, 2, \ldots, s, k \neq h. \qquad (24)\end{aligned}$$

4) Interference angle calculation based on QFT. Calculate the inner product between quantum states

$$\begin{aligned}\langle \upsilon_k''|\vartheta_h''\rangle &= \left(\mathrm{QFT}^{\otimes s-1}R_y(+\theta)^{\otimes s-1}H^{\otimes s-1}|\upsilon_k\rangle\right)^{\dagger}\\ &\quad\times\left(\mathrm{QFT}^{\otimes s-1}R_y(-\theta)^{\otimes s-1}H^{\otimes s-1}|\vartheta_h\rangle\right)\\ &k = 1, 2, \ldots, s,\ \ h = 1, 2, \ldots, s,\ \ k \neq h \qquad (25)\end{aligned}$$

where the quantum Fourier matrix $\mathrm{QFT}^{\otimes s-1}$ in $s-1$ dimension is expressed as:

$$F = \mathrm{QFT}^{\otimes s-1}. \qquad (26)$$

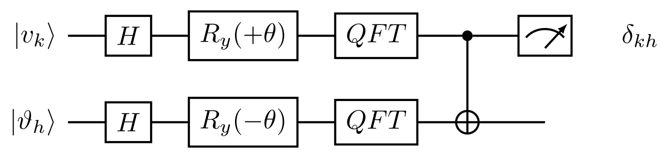

Fig. 2. Quantum circuit representation for calculating the trust path interference angle.

Therefore, (25) can be expressed as

$$\begin{aligned}\langle \upsilon_k''|\vartheta_h''\rangle &= \left(FR_y(+\theta)^{\otimes s-1}H^{\otimes s-1}|\upsilon_k\rangle\right)^{\dagger}\\ &\quad\times\left(FR_y(-\theta)^{\otimes s-1}H^{\otimes s-1}|\vartheta_h\rangle\right)\\ &k = 1, 2, \ldots, s,\ \ h = 1, 2, \ldots, s,\ \ k \neq h. \qquad (27)\end{aligned}$$

The absolute value of the inner product is calculated to represent the cosine of the interference angle

$$\cos(\delta_{kh}) = \left|\left\langle \upsilon_k'' \middle| \vartheta_h''\right\rangle\right|,\ \ k, h = 1, 2, \ldots, s,\ \ k \neq h. \qquad (28)$$

Then, the interference angle is

$$\delta_{kh} = \arccos\left(\left|\left\langle \upsilon_k'' \middle| \vartheta_h''\right\rangle\right|\right),\ \ k, h = 1, 2, \ldots, s,\ \ k \neq h. \qquad (29)$$

4) Weights Calculation for DMs: The SN, based on a trust relationship, obtains the weight of DMs, normalizes the relative importance of each DM, and then calculates the weight of DMs

$$\lambda_k = \frac{p'(d_k)}{\sum_{k=1}^{s} p'(d_k)},\ \ k = 1, 2, \ldots, s. \qquad (30)$$

C. Attribute's Weight Determination Based on TOPSIS-TODIM Method

The TOPSIS method measures the importance of objective evaluation information by calculating the distances between each alternative and the positive and negative ideal solutions, while the TODIM method characterizes the asymmetric psychological preferences of DMs under risk and gain contexts. Therefore, based on the objective evaluation concept of TOPSIS and the behavioral psychological preference representation of TODIM, this article considers both objective information and subjective perception simultaneously in the dynamic decision-making environment and constructs a dual optimization model (31), as shown at the bottom of the next page, to determine the optimal attributes' weights, as follows:

In model M_2, the objective function maximizes the sum of weighted perceptual differences $\phi(\hat{z}_{ij}^k, \hat{z}_{i'j}^k)$ among all attribute pairs, where this function, derived from TODIM theory, incorporates DMs' psychological and behavioral characteristics. The parameters $\eta, \gamma \in (0, 1)$ and $\kappa > 1$, respectively, describe their risk sensitivity and loss aversion, thereby reflecting the heterogeneity of behavioral preferences in decision-making.

Constraint condition (31-1) stipulates that the sum of the weights of all attributes is 1, and constraint condition (31-2) limits the weight of each attribute to between the minimum and maximum values of its corresponding closeness degree. Constraint condition (31-3) controls the degree of dispersion of the weights of attributes; The constraint conditions (31-4) and (31-5) are for calculating the closeness degree; Constraint conditions (31-6) are based on the TODIM method to describe the asymmetric behavioral responses of DMs in the face of

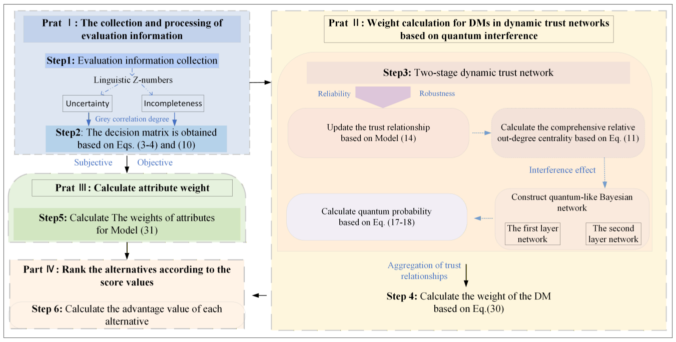

Fig. 3. Decision-making framework flowchart.

gain and loss situations. The remaining constraints impose restrictions on each parameter interval.

Theorem 4: Model M_2 has an optimal solution.

Given that the model structure is similar to Model M_1, the proof of the existence of the optimal solution can be directly referred to Theorem 3.

D. Ranking of Alternatives Based on Score Values

After calculating the weight of each DM and attribute weight, the final ranking of alternatives is obtained according to the score values

$$Q_i = \sum_{k=1}^{s} \lambda_k \sum_{j=1}^{n} \omega_j^k \hat{z}_{ij}^k, \quad i = 1, 2, \ldots, m. \tag{32}$$

E. Decision Steps

The flowchart of the proposed STN-MAQGDM model considering the interference effect under the quantum framework is shown in Fig. 3, and the specific decision-making procedure is described as follows.

Step 1: Determine the evaluation information. Each DM $d_k(k = 1, 2, \ldots, s)$ evaluates attribute $c_j(j = 1, 2, \ldots, n)$ in alternative $a_i(i = 1, 2, \ldots, m)$ and obtains the initial LZN evaluation information matrix $Z^k = (z_{ij}^k)_{m\times n}(k = 1, 2, \ldots, s)$.

Step 2: Transform the initial information evaluation matrix into a decision matrix. The evaluation information matrix is transformed into LSF matrix $\tilde{Z}^k = (\tilde{z}_{ij}^k)_{m\times n}(k = 1, 2, \ldots, s)$ according to (3) and (4), and into decision matrix $\hat{Z}^k = (\hat{z}_{ij}^k)_{m\times n}(k = 1, 2, \ldots, s)$ according to (10).

Step 3: First, the corrected trust matrix is obtained according to (11)–(14). Secondly, the QLBN is constructed based on (15)–(18), where the interference values are measured according to (19)–(29).

Step 4: DMs' weights determination based on comprehensive trust degree. Use (30) to calculate the weight of each DM $\lambda_k(k = 1, 2, \ldots, s)$.

Step 5: Determine attribute weights with TOPSIS-TODIM. On the basis of considering the internal relations and the psychology of DMs, a model (31) is constructed to solve the optimal attribute weights.

Step 6: Rank the alternatives. The (32) is used to calculate and rank the advantage values of each alternative.

IV. Application Example Analysis

A. Background

In July 2024, continuous heavy rainfall led to a major breach in the Dongting Lake area of Hunan Province, China,

$$M_2:\ \max \sum_{i=1}^{m}\sum_{i'=1}^{m}\left(\sum_{j=1}^{n}\omega_j^k \phi\left(\hat{z}_{ij}^k, \hat{z}_{i'j}^k\right)\right)$$

$$\text{s.t.}\begin{cases} \sum_{j=1}^{n}\omega_j^k = 1 & (31-1)\\ \omega_j^k \in \left[\min_i C_{ij}, \max_i C_{ij}\right] & (31-2)\\ \frac{1}{n}\sum_{j=1}^{n}\left(\omega_j^k - \frac{1}{n}\right)^2 \in \begin{bmatrix} \min_i \frac{1}{m}\sum_{i=1}^{m}\left(\frac{C_{ij}}{\sum_{i=1}^{m} C_{ij}} - \frac{1}{m}\right) \\ \max_i \frac{1}{m}\sum_{i=1}^{m}\left(\frac{C_{ij}}{\sum_{i=1}^{m} C_{ij}} - \frac{1}{m}\right)\end{bmatrix} & (31-3)\\ C_{ij} = \dfrac{\sqrt{\left(\hat{z}_{ij}^k - Z_j^{k-}\right)^2}}{\sqrt{\left(\hat{z}_{ij}^k - Z_j^{k+}\right)^2} + \sqrt{\left(\hat{z}_{ij}^k - Z_j^{k-}\right)^2}} & (31-4)\\ Z^{k+} = \left(\max_i \hat{z}_{ij}^k | j = 1, 2, \ldots, n\right) \\ Z^{k-} = \left(\min_i \hat{z}_{ij}^k | j = 1, 2, \ldots, n\right) & (31-5)\\ \phi\left(\hat{z}_{ij}, \hat{z}_{i'j}\right) = \begin{cases}\left(\hat{z}_{ij}^k - \hat{z}_{i'j}^k\right)^{\eta}, & \hat{z}_{ij}^k \geq \hat{z}_{i'j}^k \\ -\frac{1}{\kappa}\left(\hat{z}_{ij}^k - \hat{z}_{i'j}^k\right)^{\gamma}, & \hat{z}_{ij}^k < \hat{z}_{i'j}^k\end{cases} & (31-6)\\ \omega_j^k \geq 0 \quad \forall j \in 1, 2, \ldots, n & (31-7)\\ i = 1, 2, \ldots, m, \quad j = 1, 2, \ldots, n \\ i' = 1, 2, \ldots, m, \eta, \gamma \in (0, 1), \kappa > 1, \quad i \neq i' & (31-8) \end{cases} \tag{31}$$

causing severe floods, casualties, and property losses. In such an emergency disaster situation, making rapid and effective rescue decisions becomes extremely challenging. To address this challenge, the province promptly established an emergency command center composed of local authorities, fire and medical teams, transportation departments, and water experts. Considering the different viewpoints and suggestions of various units, it is crucial to adopt the MAGDM method to evaluate and select the optimal rescue alternative. Based on the actual situation of the Dongting Lake levee breach event, it is assumed that there are five DMs (d_1 to d_5). Five rescue alternatives, namely helicopter airdrop joint rescue (a_1), water–land integrated mobile rescue (a_2), key levee reinforcement and flood diversion control (a_3), local collaboration and community self-rescue coordination (a_4), and AI-assisted intelligent scheduling with uncrewed equipment rescue (a_5), they are comprehensively evaluated and selected according to five attributes: 1) speed of rescue c_1; 2) resource deployment c_2; 3) safety c_3; 4) cost-effectiveness c_4; and 5) social impact c_5.

B. Decision Steps

Step 1: Determine the evaluation information. The linguistic set A can be represented as follows: $A = \{A_0$: very poor, A_1 : poor, A_2 : slightly poor, A_3 : fair, A_4 : slightly good, A_5 : good, A_6 : very good$\}$. And the linguistic set B can be represented as follows: $B = \{B_0$: more unlikely, B_1 : unlikely, B_2 : fair, B_3 : likely, B_4 : more likely$\}$. On this basis, the initial evaluation information of each DM is collected and the evaluation information matrix $Z^k = (z_{ij}^k)_{5\times5} (k = 1, 2, \dots, 5)$ is obtained, as follows: $Z^1 - Z^5$, as shown at the bottom of page.

Step 2: LZNs can be converted into crisp numbers according to (3) and (4), and a decision matrix $\hat{Z}^k = (\hat{z}_{ij}^k)_{5\times5} (k = 1, 2, \dots, 5)$ with crisp number representation can be obtained through (10).

Step 3: First, the initial trust matrix of the trust network is $T = (t_{ij})_{5\times5}$, this article assumes that $\rho_4 = 10$, $\rho_5 = \rho_6 = 0.5$ and obtains the two-stage trust matrix $T' = (t'_{ij})_{5\times5}$ by solving model (14)

$$T = \begin{bmatrix} - & 0.5 & 0.4 & 0.7 & - \\ 0.2 & - & - & 0.6 & 0.8 \\ 0.3 & - & - & 0.4 & - \\ 0.4 & - & - & - & 0.6 \\ 0.8 & 0.9 & - & 0.4 & - \end{bmatrix}$$

$$T' = \begin{bmatrix} - & 0.5953 & 0.4364 & 0.7964 & 0.1063 \\ 0.3275 & - & 0.0632 & 0.6964 & 0.9261 \\ 0.3875 & 0.0021 & - & 0.4491 & 0.0105 \\ 0.5785 & 0.1066 & 0.1906 & - & 0.7481 \\ 0.8663 & 0.9909 & 0.0728 & 0.5000 & - \end{bmatrix}.$$

On this basis, the comprehensive relative out-degree centrality of each DM is calculated

$$\mathrm{COD}_1 = 0.9685, \quad \mathrm{COD}_2 = 0.8482, \quad \mathrm{COD}_3 = 0.3405$$
$$\mathrm{COD}_4 = 0.9433, \quad \mathrm{COD}_5 = 1.0000.$$

$$Z^1 = \begin{pmatrix} (A_3, B_2) & (A_6, B_3) & (A_5, B_1) & (A_5, B_1) & (A_4, B_3) \\ (A_4, B_4) & (A_4, B_2) & (A_4, B_2) & (A_5, B_2) & (A_0, B_4) \\ (A_1, B_4) & (A_3, B_3) & (A_3, B_3) & (A_3, B_2) & (A_3, B_2) \\ (A_4, B_3) & (A_2, B_1) & (A_6, B_4) & (A_6, B_2) & (A_4, B_4) \\ (A_5, B_1) & (A_2, B_3) & (A_5, B_3) & (A_0, B_0) & (A_3, B_3) \end{pmatrix}$$

$$Z^2 = \begin{pmatrix} (A_2, B_2) & (A_4, B_1) & (A_3, B_2) & (A_3, B_2) & (A_2, B_4) \\ (A_3, B_1) & (A_5, B_4) & (A_2, B_3) & (A_2, B_4) & (A_3, B_2) \\ (A_4, B_3) & (A_3, B_3) & (A_6, B_4) & (A_4, B_1) & (A_4, B_3) \\ (A_3, B_4) & (A_2, B_4) & (A_4, B_1) & (A_4, B_3) & (A_5, B_1) \\ (A_6, B_2) & (A_0, B_0) & (A_6, B_2) & (A_3, B_3) & (A_3, B_2) \end{pmatrix}$$

$$Z^3 = \begin{pmatrix} (A_6, B_4) & (A_6, B_3) & (A_2, B_3) & (A_2, B_3) & (A_4, B_0) \\ (A_3, B_4) & (A_5, B_4) & (A_1, B_1) & (A_3, B_4) & (A_1, B_1) \\ (A_2, B_2) & (A_4, B_2) & (A_0, B_2) & (A_1, B_2) & (A_2, B_2) \\ (A_4, B_3) & (A_2, B_2) & (A_3, B_3) & (A_5, B_1) & (A_1, B_3) \\ (A_4, B_1) & (A_3, B_3) & (A_5, B_4) & (A_6, B_3) & (A_5, B_4) \end{pmatrix}.$$

$$Z^4 = \begin{pmatrix} (A_3, B_0) & (A_4, B_3) & (A_4, B_3) & (A_4, B_2) & (A_3, B_4) \\ (A_5, B_3) & (A_3, B_4) & (A_3, B_0) & (A_3, B_3) & (A_2, B_3) \\ (A_4, B_4) & (A_4, B_2) & (A_1, B_1) & (A_2, B_4) & (A_4, B_2) \\ (A_6, B_2) & (A_2, B_1) & (A_4, B_2) & (A_5, B_3) & (A_3, B_1) \\ (A_2, B_1) & (A_0, B_0) & (A_6, B_4) & (A_3, B_0) & (A_4, B_0) \end{pmatrix}$$

$$Z^5 = \begin{pmatrix} (A_4, B_2) & (A_3, B_4) & (A_5, B_1) & (A_4, B_1) & (A_6, B_0) \\ (A_6, B_3) & (A_3, B_3) & (A_4, B_2) & (A_3, B_2) & (A_4, B_1) \\ (A_5, B_2) & (A_5, B_2) & (A_0, B_4) & (A_2, B_3) & (A_3, B_3) \\ (A_0, B_2) & (A_2, B_3) & (A_2, B_3) & (A_0, B_0) & (A_5, B_2) \\ (A_1, B_1) & (A_6, B_4) & (A_3, B_2) & (A_1, B_3) & (A_5, B_4) \end{pmatrix}.$$

Second, let the rotation angle $\theta = 60°$, and calculate the number of interference angles based on quantum gate transformation according to (20)–(29)

$$\delta_{12} = 33.8964°,\quad \delta_{13} = 52.8169°,\quad \delta_{14} = 32.5721°$$
$$\delta_{15} = 40.7958°,\quad \delta_{23} = 51.4317°,\quad \delta_{24} = 45.5263°$$
$$\delta_{25} = 54.3596°,\quad \delta_{34} = 32.5096°,\quad \delta_{35} = 73.8878°$$
$$\delta_{45} = 42.2918°.$$

Finally, the comprehensive confidence of each DM considering quantum interference is calculated according to (15)–(18)

$$p'(d_1) = 0.2622,\quad p'(d_2) = 0.3036,\quad p'(d_3) = 0.1025$$
$$p'(d_4) = 0.2351,\quad p'(d_5) = 0.3661.$$

Step 4: Calculate the weight of each DM according to (30)

$$\lambda_1 = 0.2357,\quad \lambda_2 = 0.2192,\quad \lambda_3 = 0.0712$$
$$\lambda_4 = 0.2121,\quad \lambda_5 = 0.2618.$$

Step 5: The parameter η and γ are taken as 0.88 and κ is taken as 2.25 [45]. Solve the model (31) to obtain the optimal attributes' weights evaluated by five different DMs

$$W^1 = (0.3712, 0.1877, 0.0404, 0.1294, 0.2713)$$
$$W^2 = (0.1232, 0.0480, 0.1697, 0.3642, 0.2949)$$
$$W^3 = (0.0170, 0.2687, 0.3207, 0.2747, 0.1189)$$
$$W^4 = (0.1876, 0.4873, 0.7567, 0.0920, 0.0764)$$
$$W^5 = (0.0317, 0.0158, 0.5747, 0.3766, 0.0012).$$

Step 6: Calculate the score value of each alternative using (32)

$$Q_1 = 0.5752,\quad Q_2 = 0.6232,\quad Q_3 = 0.5964$$
$$Q_4 = 0.5676,\quad Q_5 = 0.5437.$$

Therefore, the various alternatives are ranked according to the score values, and the ranking result is: $a_2 \succ a_3 \succ a_1 \succ a_4 \succ a_5$.

C. Sensitivity Analysis

In this section, sensitivity analysis of some parameters in the model will be carried out to observe their influence on the quantum decision model proposed in this article.

1) Sensitivity analysis of rotation angle θ

The rotation angle regulates the superposition and interference effects between quantum states, thus affecting the DM's final decision. Fig. 4 shows the variation of each interference angle at rotation angle $\theta \in [0, \pi]$.

Fig. 4 shows that all curves exhibit extrema around $\theta \approx 90°$ and return to their initial states at $\theta = 0°$ and $\theta = 180°$, showing clear symmetry and periodicity. As θ increases from 0° to 90°, most interference angles δ_{kh} gradually increase, indicating that the expanding phase difference between paths enhances the interference effect. When θ exceeds 90°, δ_{kh} decreases correspondingly, suggesting that the interference weakens. The response intensity varies among different pairs

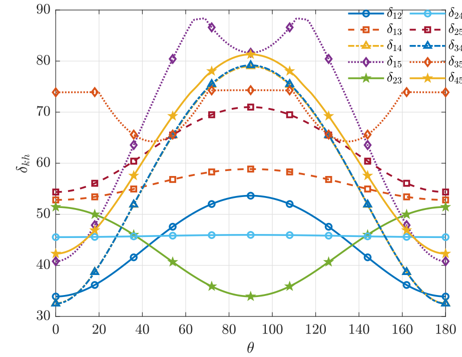

Fig. 4. Influence of rotation angle θ variation on interference angle $\delta_{kh}(k \neq h = 1, 2, \ldots, 5)$.

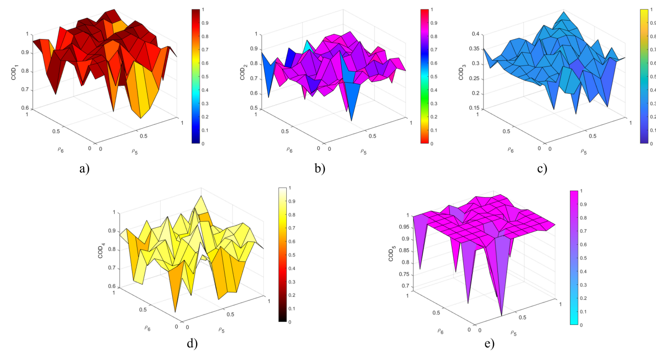

Fig. 5. Influence of discount factor ρ_5 and adjustment parameter ρ_6 on $\text{COD}_h(h = 1, 2, \ldots, 5)$. (a) COD of d_1. (b) COD of d_2. (c) COD of d_3. (d) COD of d_4. (e) COD of d_5.

of DMs: δ_{14}, δ_{15}, and δ_{34} exhibit larger fluctuations, showing strong interference sensitivity to phase changes, while δ_{13} and δ_{24} vary more smoothly, implying weaker interference.

Therefore, in the process of quantum decision-making, the setting of the rotation angle has an important impact on the decision result. A reasonable selection of the rotation angle can effectively control decision interference and improve decision quality.

2) Sensitivity analysis of discount factor ρ_5 and adjustment parameter ρ_6

In the trust network model, discount factor ρ_5 and adjustment parameter ρ_6 are two key parameters, which directly affect the transfer and attenuation of trust values between nodes, and thus affect the trust distribution of the whole network. Through the sensitivity analysis of these two parameters, the influence of their changes on the comprehensive relative out-degree centrality of each DM can be observed from Fig. 5.

As can be seen from Fig. 5, a larger ρ_5 (lower attenuation) generally increases the trust index of each node, because trust decreases less in the transmission process, and indirect trust has a greater impact. A larger ρ_6 (strong indirect trust influence) will increase the trust index, because the weight of indirect trust increases and the trust relationship between nodes becomes more complex. Therefore, the influence of the two parameters on the trust network is analyzed from four perspectives as follows.

1) When both ρ_5 and ρ_6 are large, the decay of trust transmission is slow, and the influence of indirect trust is

TABLE I
COMPARISON OF THE PROPOSED METHOD AND PREVIOUS DECISION MODELS

Models	Information type	Information representation	Information uncertainty type	Information reliability description	DMs' weights	Trust relationships	Attributes' weights	Interference term calculation
Zha et al. [36]	Crisp number	-	-	-	Known	No	Known	-
Zhang et al. [3]	2-Tuple Linguistic Representation Model	Linguistic term + deviation value	Fuzziness	Unable to describe	Known	No	Known	-
Wu and Liao [47]	Probabilistic linguistic term set	Multiple linguistic terms +probability distribution	Fuzziness + randomness	Indirectly expressed	-	No	Unknown	-
Wu et al. [32]	Linguistic term set	Single linguistic term	Fuzziness	-	Unknown	No	Unknown	Constant value
She et al. [34]	Crisp number	-	-	-	-	No	Unknown	Deng entropy
Wang et al. [29]	Crisp number	-	-	-	Unknown	Yes	Unknown	Similarity heuristic approach
Zhang et al. [47]	Linguistic distribution assessment	Linguistic distribution information	Fuzziness + randomness	Indirectly expressed	Unknown	No	Known	Optimization modeling solution
The proposed model	LZN	Linguistic term + reliability linguistic term	Fuzziness + reliability	Clearly described	Unknown	Yes	Unknown	Quantum gate

substantial, resulting in a very close trust relationship between nodes in the trust network and a high overall trust degree. This combination is suitable for decision-making environments that require high levels of trust and complex trust relationships.

2) When both ρ_5 and ρ_6 are small, the decay of trust transmission is fast, and the influence of indirect trust is minimal, resulting in a loose trust relationship between nodes in the trust network and a low overall trust degree. This combination is appropriate for environments that emphasize independent decision-making and direct trust relationships.

3) When ρ_5 is large and ρ_6 is small, the decay of trust transmission is slow, but the influence of indirect trust is minimal, resulting in the trust network primarily relying on direct trust relationships. This combination is suitable for environments where direct trust needs to be maintained while giving due consideration to the transfer of trust.

4) When ρ_5 is small and ρ_6 is large, the decay of trust is fast in the process of trust transmission, but the influence of indirect trust is substantial, resulting in complex but not close trust relationships in the trust network. This combination is suitable for environments that need to simplify the process of trust transfer but still value the indirect effects of trust.

In summary, by analyzing different combinations of discount factors ρ_5 and adjustment parameters ρ_6, the trust transmission and attenuation mechanism in the SN-GDM can be better optimized, thereby improving the accuracy and effectiveness of decision-making.

D. Comparative Analysis

1) Qualitative Comparative Analysis: In order to verify the superiority and effectiveness of the model in this article, the model proposed in this article is qualitatively compared with some existing models, as shown in Table I.

As shown in Table I, the existing MAGDM models still exhibit evident limitations in terms of theoretical structure and information representation. Traditional models mainly focus on constructing DMs' weights, attribute weights, and social relationships, while neglecting the mutual interference effects among DMs, which makes it difficult to reflect the genuine group interaction mechanism. Although some quantum decision models introduce quantum superposition and interference mechanisms, they usually rely on subjective assignment in weight determination or fail to fully consider the dynamic transmission of trust relationships, leading to insufficient coupling between quantum effects and social structures. In addition, the existing interference angle measurement methods mainly focus on vector directions or internal chaotic states, making it difficult to effectively characterize nonlinear dependencies and complex phase interactions.

In terms of information representation, traditional linguistic sets and two-tuple linguistic numbers can only describe the fuzziness of evaluations, without capturing the reliability of decision information. Although probabilistic linguistic term sets can express the distributional characteristics of multiple linguistic terms, their probabilities merely describe the range of uncertainty and fail to distinguish the degree of information reliability. In contrast, LZNs employ a dual-layer structure of "linguistic term + reliability linguistic term," which simultaneously characterizes fuzziness and reliability, thereby achieving a hierarchical representation of decision information.

In summary, the proposed method realizes adaptive updating of trust values within a dynamic trust network and determines DMs' weights based on the interference effects among trust paths. The interference angle is calculated through quantum gate transformations to capture nonlinear dependencies. Furthermore, by integrating the dual optimization mechanism of the TOPSIS–TODIM method, the model dynamically determines attribute weights. With the high-dimensional expression and reliability modeling capability of LZNs, the proposed framework demonstrates enhanced flexibility and robustness in complex and uncertain decision-making environments.

2) Quantitative Comparative Analysis: In order to verify the effectiveness of the proposed method, several existing decision models are compared and analyzed, and the final ranking results obtained by different methods are shown in Table II. Although these models differ in theoretical mechanisms, they share a common goal of ranking alternatives and produce structurally comparable output. To ensure comparability, we adopt a unified input-output design: the same decision matrices are used as inputs for all models, and their respective ranking results are analyzed.

It can be seen from Table II that compared with the traditional decision model, the alternative scheme ranking obtained by the proposed method is significantly different. For example, the optimal alternative obtained by TOPSIS [48] method is a_4, and the optimal alternative obtained by TODIM [49] method is a_3, while the optimal alternative obtained by this article is a_2. The reason for this difference is that the traditional model does not consider the interference effect between DMs. As a result, the ranking of alternatives is subject to subjective influence among DMs.

In contrast, when compared with the Quantum model [32], the optimal choice is a_2, but there is a big difference between the ordering of other schemes and the results in this article.

TABLE II
COMPARISON OF RANKING RESULTS OF DIFFERENT MODELS

Methods	Ranking indices	Final ranking orders
TOPSIS [48]	$C_1^*=0.6204,\ C_2^*=0.4415,\ C_3^*=0.6576,\ C_4^*=0.8761,\ C_5^*=0.5281.$	$a_4 \succ a_3 \succ a_1 \succ a_5 \succ a_2$
TODIM [49]	$\phi(x_1)=-0.7355,\ \phi(x_2)=-3.8121,\ \phi(x_3)=-0.8432,\ \phi(x_4)=1.0140,\ \phi(x_5)=-1.8086.$	$a_3 \succ a_4 \succ a_5 \succ a_2 \succ a_1$
Quantum model [32]	$p(a_1)=0.2044,\ p(a_2)=0.2807,\ p(a_3)=0.1736,\ p(a_4)=0.1856,\ p(a_5)=0.1557.$	$a_2 \succ a_1 \succ a_4 \succ a_3 \succ a_5$
Quantum model [34]	$p(a_1)=0.1502,\ p(a_2)=0.3590,\ p(a_3)=0.1950,\ p(a_4)=0.1388,\ p(a_5)=0.1569.$	$a_2 \succ a_3 \succ a_5 \succ a_1 \succ a_4$
Quantum model [29]	$p(a_1)=0.2706,\ p(a_2)=0.2435,\ p(a_3)=0.0745,\ p(a_4)=0.2319,\ p(a_5)=0.1796.$	$a_1 \succ a_2 \succ a_4 \succ a_3 \succ a_5$
The proposed method	$Q_1=0.5752,\ Q_2=0.6232,\ Q_3=0.5964,\ Q_4=0.5676,\ Q_5=0.5437.$	$a_2 \succ a_3 \succ a_1 \succ a_4 \succ a_5$

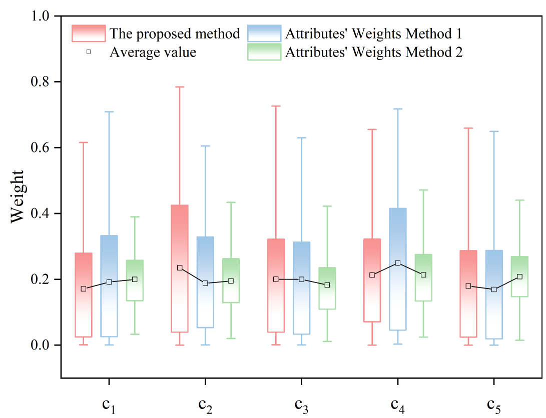

Fig. 6. Comparison of the weight distribution of three attribute weight calculation methods.

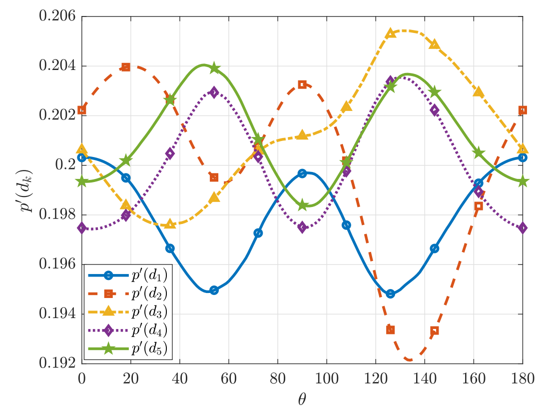

Fig. 7. Variation of comprehensive trust level $p'(d_k)$ with rotation angle θ under randomized trust relationships.

This is because the model cannot measure the specific angle of the phase angle, which affects the accuracy of the results. Quantum model [34] considers a static decision process without considering the influence of communication between DMs on the decision process, and the interference value of the Deng entropy measure only considers the chaotic state inside the system. The optimal alternative obtained by the Quantum model [29] is a_1, but this model only considers the direction of the vector through the cosine similarity measurement of the interference angle, and does not fully consider the interference of the phase information. The method proposed in this article uses quantum gate transform and QFT to measure the interference angle, fully considers the phase information, updates the DM's weight in combination with the dynamic trust transfer process, and considers the DM's risk attitude when calculating attributes' weights. Therefore, the results of this model are more consistent with the real complex situation.

E. Simulation Analysis

1) Comparative Analysis of the Weighting Methods of Attributes: To verify the effectiveness and superiority of the proposed attributes' weight optimization model in MAGDM, two representative methods are selected for comparison: Attributes' Weights Method 1 [32] (entropy weight method) and Attributes' Weights Method 2 [50] (an optimization model based on the idea of deviation maximization). A 5×5 decision matrix is constructed, with elements randomly generated within the [0, 1] interval using the MATLAB "rand" function, to simulate the fluctuation characteristics of attribute evaluations under different decision scenarios. Each method is independently executed 100 times to enhance the robustness of the results. The outcomes are shown in Fig. 6.

As illustrated in Fig. 6, the attributes' weights obtained by Attributes' Weights Method 1 [32] and Attributes' Weights Method 2 [50] are generally concentrated with low variability, and some attributes yield nearly identical weights across multiple runs. This indicates that the two traditional methods have limited responsiveness to attribute importance differences in uncertain and complex environments. Attributes' Weights Method 1 [32], based on information entropy, relies heavily on the dispersion of attribute values but tends to overlook real contribution differences; Attributes' Weights Method 2 [50] enhances discrimination by maximizing deviation but lacks consideration of DMs' psychological perceptions and subjective preferences. In contrast, the proposed method demonstrates a wider range of weight variation and more significant differentiation across attributes, indicating a stronger ability to identify key attributes under uncertainty.

2) Simulation Analysis of the Impact of Rotation Angle on Comprehensive Trust Level: To verify the influence of the rotation angle θ on the comprehensive trust probability $p'(d_k)$, a numerical simulation was conducted under the framework of SN trust propagation. In this simulation, the prior probabilities were uniformly distributed ($p(d_h) = 1/s$), and the trust relationships t'_{hk} were randomly generated within $[0, 1]$ and calculated by (16) to obtain the conditional probabilities. To reduce randomness and ensure robustness, the experiment was repeated 1000 times and averaged, with the results shown in Fig. 7.

Overall, Fig. 7 indicates that as θ increases from 0° to 180°, the comprehensive trust values $p'(d_k)$ of different DMs exhibit slight but distinguishable periodic fluctuations. More specifically, $p'(d_2)$ and $p'(d_1)$ reach local minima around

$\theta \in [120°, 140°]$, whereas $p'(d_3)$ and $p'(d_5)$ attain peaks in the same region, reflecting alternating enhancement effects among trust propagation paths under phase modulation. In conclusion, the variation of θ enables dynamic fine-tuning of the interference intensity between trust paths, thereby improving the adaptability and robustness of trust propagation in complex SN decision environments.

V. Conclusion

To address the problems of information uncertainty, time sensitivity, and dynamic interference existing in uncertain decisions, this article proposes a quantum group decision-making model of SN-GDM, which effectively improves the efficiency and response speed of such uncertain decisions. First, complex and uncertain information is expressed by using LZNs, and it is transformed into clarity numbers through the gray relational degree method, significantly improving the clarity and efficiency of information processing. Second, the dynamic two-stage SN structure was considered to update the degree of trust among DMs. On this basis, the interference effect between trust paths was measured based on the quantum gate transform and QFT. Furthermore, the attribute weight optimization model constructed in combination with the TOPSIS-TODIM idea dynamically determines the attributes' weights on the basis of integrating the laws of objective data and the subjective preferences of DMs, reducing the time loss and subjective deviation caused by human intervention.

Therefore, the method proposed in this article as a whole has a modular feature, covering steps such as trust weight calculation, attribute weight optimization, and alternative solution ranking. It can not only be rapidly deployed and flexibly applied in the context of emergency scenario decision-making, but also be extended to SN-GDM problems with uncertainty and dynamic interaction characteristics. For instance, in scenarios such as risk assessment, supplier selection, and conflict coordination, the proposed method can more accurately depict the trust evolution mechanism and behavioral differences among DMs through the integration of dynamic trust transfer and quantum interference effects.

This article overcomes the limitation of traditional methods that regard DMs as independent individuals and expands the measurement method of quantum interference angles in the field of SN-GDM. However, the number of interference angles is calculated using the inner product between quantum states, which does not fully consider the influence of negative interference effects between DMs. Furthermore, as the number of DMs and attributes increases, the computational complexity of the model increases sharply. To address this issue, future research will explore how to effectively incorporate machine learning into multiattribute quantum group decision processes and deepen quantum gate transformations to measure interference values more precisely.

Appendix A
Proof of Theorem 1

1) When the LSF is $F_1^*(\alpha)$, three properties of Theorem 1 are satisfied.

1) $F_1^*(0) = (\ln 1)/(2\ln 3) = 0, F_1^*(m) = (\ln 3)/(2\ln 3) = 1/2, F_1^*(2m) = 1 - (\ln 1)/(2\ln 3) = 1.$ □

2) When $\alpha = 1, 2, \ldots, m$,

$$\frac{d}{d\alpha}F_1^*(\alpha) = \frac{1}{2\ln 3}\cdot\frac{\beta_1\left(\frac{\alpha}{m}\right)^{\beta_1-1} + \beta_2\left(\frac{\alpha}{m}\right)^{\beta_2-1}}{m\left(1 + \left(\frac{\alpha}{m}\right)^{\beta_1} + \left(\frac{\alpha}{m}\right)^{\beta_2}\right)} \geq 0.$$

When $\alpha = m + 1, m + 2, \ldots, 2m$,

$$\frac{d}{d\alpha}F_1^*(\alpha) = \frac{\beta_1\left(\frac{2m-\alpha}{m}\right)^{\beta_1-1} + \beta_2\left(\frac{2m-\alpha}{m}\right)^{\beta_2-1}}{2\ln 3\cdot m\left(1 + \left(\frac{2m-\alpha}{m}\right)^{\beta_1} + \left(\frac{2m-\alpha}{m}\right)^{\beta_2}\right)} \geq 0.$$

Since the function at $\alpha = 0, 1, 2, \ldots, 2m$ is continuous, so that $F_1^*(\alpha)$ is $\alpha = 0, 1, 2, \ldots, 2m$ is monotonically increasing. □

3)

$$\begin{aligned}
&F_1^*(i) + F_1^*(j)\\
&= \frac{\ln\left(1 + \left(\frac{i}{m}\right)^{\beta_1} + \left(\frac{i}{m}\right)^{\beta_2}\right)}{2\ln 3}\\
&\quad + 1 - \frac{\ln\left(1 + \left(\frac{2m-j}{m}\right)^{\beta_1} + \left(\frac{2m-j}{m}\right)^{\beta_2}\right)}{2\ln 3}\\
&\overset{j=2m-i}{=} \frac{\ln\left(1 + \left(\frac{i}{m}\right)^{\beta_1} + \left(\frac{i}{m}\right)^{\beta_2}\right)}{2\ln 3}\\
&\quad + 1 - \frac{\ln\left(1 + \left(\frac{i}{m}\right)^{\beta_1} + \left(\frac{i}{m}\right)^{\beta_2}\right)}{2\ln 3} = 1.
\end{aligned}$$

□

2) When the LSF is $F_2^*(\alpha)$, three properties of Theorem 1 are satisfied.

1) $F_2^*(0) = 0, F_2^*(m) = 1/2(1 - \cos((\pi m)/(2m))) = 1/2, F_2^*(2m) = 1/2(1 - \cos(\pi)) = 1.$ □

2) $d/(d\theta)F_2^*(\theta) = (\pi)/(4m)\sin((\pi a)/(2m))$, since $\sin((\pi a)/(2m))$ is in $\alpha = 0, 1, 2, \ldots, 2m$ is positive,so $(\pi)/(4m)\sin((\pi a)/(2m)) > 0$,that is $F_2^*(\alpha)$ in $\alpha = 0, 1, 2, \ldots, 2m$ is monotonically increasing. □

3) $F_2^*(i) + F_2^*(j) = 1/2(1 - \cos((\pi i)/(2m))) + 1/2(1 - \cos((\pi j)/(2m)))$, since $\cos((i\pi)/(2m)) = -\cos((\pi(2m - i))/(2m))$, so $F_2^*(i) + F_2^*(j) = 1/2(1 - \cos((\pi i)/(2m))) + 1/2(1 - \cos((\pi(2m - i))/(2m))) = 1/2(1 - \cos((\pi i)/(2m))) + 1/2(1 + \cos((\pi i)/(2m))) = 1.$ □

Appendix B
Proof of Theorem 2

1) Since $\min\limits_k \min\limits_i d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+}) \leq d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+}) \leq \max\limits_k \max\limits_i d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+})$ and $\rho_3 > 0$, $\min\limits_k \min\limits_i d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+}) + \rho_3 \max\limits_k \max\limits_i d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+}) \leq d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+}) + \rho_3 \max\limits_k \max\limits_i d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+})$, which is: $0 < \tilde{z}_{ij}^k \leq 1$. □

2) When $d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+}) < d(\tilde{z}_{mn}^k, \tilde{z}_0^{k+})$, then

$$\begin{aligned}
&\frac{\min_k \min_i d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+}) + \rho_3 \max_k \max_i d(\tilde{z}_{mn}^k, \tilde{z}_0^{k+})}{d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+}) + \rho_3 \max_k \max_i d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+})}\\
&> \frac{\min_k \min_i d(\tilde{z}_{mn}^k, \tilde{z}_0^{k+}) + \rho_3 \max_k \max_i d(\tilde{z}_{mn}^k, \tilde{z}_0^{k+})}{d(\tilde{z}_{mn}^k, \tilde{z}_0^{k+}) + \rho_3 \max_k \max_i d(\tilde{z}_{mn}^k, \tilde{z}_0^{k+})}
\end{aligned}$$

that is $\hat{z}_{ij}^k > \hat{z}_{mn}^k$. □

3) The partial derivative of ρ_3 according to (10) is as follows:

$$\frac{\partial \hat{z}_{ij}^k}{\partial \rho_3} = \frac{\left[\max_k \max_i d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+})\right]\left(d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+}) - \min_k \min_i d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+})\right)}{\left(d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+}) + \rho_3 \max_k \max_i d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+})\right)^2}.$$

Since $\min_k \min_i d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+}) \leq d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+})$, then $d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+}) - \min_k \min_i d(\tilde{z}_{ij}^k, \tilde{z}_0^{k+}) \geq 0$, which is $\frac{\partial \hat{z}_{ij}^k}{\partial \rho_3} \geq 0$, so $\hat{z}_{ij}^k$ is monotonically increasing at $\rho_3 \in (0, 1)$. □

Appendix C
Proof of Theorem 3

The objective function is continuous, since it is composed of linear terms and a hinge–squared penalty. The constraint set is closed and bounded because $0 \leq t'_{kh} \leq 1$ and all remaining constraints are linear. For all (k, h), $\varepsilon_2 \geq \rho_5\rho_6(s-1)$, under this condition, the feasible set is nonempty. Hence, by the Weierstrass theorem, a continuous objective over a nonempty compact feasible set attains its maximum; therefore, the model admits an optimal solution. □

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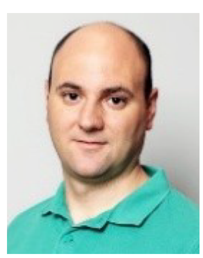

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